Q1.

Two pupils investigated the effect of temperature on how fast oil flows through a funnel. They used the equipment in the photograph below.



(a) They measured the time taken for all the oil to flow through the funnel.

What equipment did they use to measure the time?

.....

1 mark

(b) Complete the table below to show what they should do with each factor in their investigation.
Tick **one** box for each factor.

factor	change it	keep it the same	measure it
temperature of the oil			
type of oil			
volume of oil			
time taken for all the oil to flow through the funnel			

2 marks

(c) (i) Look at their results in the table below.

temperature of oil (°C)	time taken for all the oil to flow through the funnel (s)
22	131
40	35
60	22
80	19

What happens to the time taken for the oil to flow through the funnel as its temperature increases?

.....

.....

1 mark

(ii) How long would it take for all the oil to flow through the funnel at 15°C?

Choose from the following times.
Tick the correct box.

15 seconds ☐

80 seconds ☐

131 seconds ☐

180 seconds ☐

1 mark
maximum 5 marks

Q1.

(a) any **one** from

- stopclock
  'clock' is insufficient
- stopwatch
  'watch' is insufficient
  accept 'timer'

1 (L3)

(b)

factor	change it	keep it the same	measure it
temperature of the oil	✓		
type of oil		✓	
volume of oil		✓	
time taken for all the oil to flow through the funnel			✓

if all four answers are correct, award two marks
if three answers are correct, award one mark
if more than one box is ticked in any row, count the answer as incorrect for that row

2 (L4)

(c) (i) any **one** from

- it **or** time gets less
  *accept 'the time gets faster' **or** 'it gets faster'*
  *do **not** accept 'the time gets slower' unless linked to a decrease in temperature*
- it **or** time decreases
  accept 'the oil flows faster'
  accept 'it goes through faster'

1 (L4)

(ii) 180 seconds ✓

if more than one box is ticked, award no mark

1 (L3)

[5]